

Massage & manual therapy

KRKT:TIDE · KRKT's Type 1 Diabetes Education

In an emergency. Someone with type 1 who is confused, very drowsy, can't swallow or is unconscious: call 911.

Education, not medical advice. Every dose, ratio and target belongs with your own diabetes team.

Massage & manual therapy

What hands-on therapy can and can't do with type 1, the few things that change on the table, and what to tell the therapist first.

This section is for people with type 1 diabetes who are thinking about massage therapy, orthotherapy or other hands-on care — and for the massage therapists and orthotherapists who treat them.

Start with the reassurance: most people with type 1 can book a massage like anyone else. What changes is a short list of practical things. Massage, heat and working muscles can make a recent insulin dose act faster. A CGM sensor can read falsely low when it is pressed on, and oil stops its adhesive holding. Numb feet can't tell a therapist when the pressure or the heat is too much. And in Québec, choosing a therapist works differently from the rest of Canada, because neither massage therapy nor orthotherapy is a regulated profession here.

The pages below cover who does what and what massage can honestly claim, insulin and lows on the table, sensors and pumps, what changes with complications, what to tell the therapist, and a page written for therapists themselves. There is also a [printable card](#) to hand over at the first visit.

This is education, not medical advice. Where to inject, when, and how long to keep an area clear belong with your diabetes team.

Sources: Diabète Québec, « Spa, sauna et massothérapie » · Fédération québécoise des massothérapeutes (FQM) · FIT Canada 2020 · Larkin et al., Diabetes Care 2014 (DCCT/EDIC)

Who does what, and what massage can honestly claim

Massage therapists, orthotherapists, kinesiologists and physiotherapists in Québec, why people with type 1 book, and what the evidence supports.

A lot of what circulates online about massage and diabetes promises more than the research shows. This page sorts out who the different hands-on therapists are in Québec, why people with type 1 often end up seeing one, and which benefits actually hold up.

Four titles on the clinic door

Massage therapist (massothérapeute). Therapeutic massage. Massage therapy is **not regulated** in Québec: there is no professional order, and the title isn't protected. Standards come from the association the therapist belongs to. Ontario and British Columbia regulate massage therapists; Québec does not.

Orthotherapist (orthothérapeute). Massage combined with movement therapy, aimed at muscle pain and joint stiffness. It is a specialisation taken after massage training — roughly 1,000 to 1,300 hours in all, depending on the school. Orthotherapy is **not regulated** either.

Kinesiologist (kinésiologue). Exercise as treatment, represented by a federation (FKQ). Québec announced in June 2026 that kinesiology will join a professional order, but that is not in force yet.

Physiotherapist (physiothérapeute). **Regulated** by the Ordre professionnel de la physiothérapie du Québec (OPPQ), which keeps a public directory. The first call for an injury, a frozen shoulder, or anything that needs a diagnosis.

In Québec, a *kinésithérapeute* is an unregulated movement therapist. It is not the same as the French *masseur-kinésithérapeute*, which in France is the regulated physiotherapy equivalent. People who have lived in France often mix the two up.

Why people with type 1 end up on the table

Hand and shoulder problems are common after many years with type 1, and they are often mistaken for ordinary ageing.

66%

of 1,217 adults in the DCCT/EDIC study — on average 31 years with type 1 — had at least one hand or shoulder condition

31%

frozen shoulder: a painful shoulder that slowly stiffens

30%

carpal tunnel: numb, tingling fingers, often at night

28%

trigger finger: a finger that catches or locks when it bends

22%

stiff hands — unable to press the palms flat together (the "prayer sign")

Dupuytren's contracture affected 9%. Frozen shoulder is about five times more likely with diabetes than without.

These figures come from people who were diagnosed decades ago, and how common these conditions are with today's care isn't known yet. The leading explanation is that glucose slowly stiffens collagen over years (glycation) — a matter of time and exposure, not a verdict on effort. In the original trial, people who had received intensive treatment and those who had received conventional treatment had similar rates: 64% versus 68%, not a significant difference.

What massage can honestly claim

Reasonably supported. Feeling calmer. A meta-analysis of 37 trials found that a single massage session lowered anxiety, blood pressure and heart rate. For frozen shoulder in people with diabetes, joint mobilisation combined with exercise had the largest effect across 8 trials — although the evidence is low quality.

Disputed. "Massage lowers stress hormones." One review reported cortisol falling by about a third after massage. A stricter meta-analysis, comparing massage groups with control groups, found the effect on cortisol indistinguishable from zero. And stress itself moves blood sugar differently in different people with type 1: up for some, down for others, and not at all for some.

Not shown. Treating diabetes. A few small studies — mostly in children, mostly without a proper comparison group — saw glucose fall after massage sessions. The most likely reason is that massage speeds up the absorption of insulin already injected. That is a reason to check your glucose, not a therapy. There is also no good evidence that massage prevents or treats complications by "improving circulation", and no diabetes guideline lists it.

Evidence grade

Every claim about massage and blood sugar rests on low-quality evidence: small, short studies, most of them without a control group, several using older insulins.

So go because it feels good, eases tension and helps you move — not to treat diabetes. That is a perfectly good reason to book.

Sources: Office québécois de la langue française, Grand dictionnaire terminologique · Mon Réseau Plus (RMPQ) · RITMA · Gouvernement du Québec, announcement of 16 June 2026 · Larkin et al., Diabetes Care 2014 (DCCT/EDIC) · Zreik et al., Muscles Ligaments Tendons Journal 2016 · Moyer et al., Psychological Bulletin 2004 · Moyer et al., Journal of Bodywork and Movement Therapies 2011 · Field et al., International Journal of Neuroscience 2005 · Alsubheen et al., Archives of Physical Medicine and Rehabilitation 2019 · Ezzo et al., Diabetes Spectrum 2001 · Sajedi et al., Acta Medica Iranica 2011 · Wiesli et al., Diabetes Care 2005

Insulin sites, heat and lows on the table

Why massage, heat and working muscles make a recent insulin dose act faster, and how to check, prepare and treat a low during a session.

This is the part of massage that is genuinely different with type 1. Rubbing, heating or working the muscle under a recent injection gets insulin into the blood faster — and a session is also a setting where a low is easy to miss. This page explains both, and what to do about them.

Insulin moves faster under a working hand

This effect has been measured since the late 1970s.

~6x

faster insulin absorption when an injection site was massaged for 30 minutes — and blood sugar fell faster too

110%

more absorption after two sauna sessions, with blood sugar 3.0 to 3.3 mmol/L lower

135%

more absorption in the first ten minutes of leg exercise after a leg injection

Hot stones, heat packs and hot tubs work the same way as a sauna: heat speeds absorption.

These were small studies — nine volunteers in the main massage study — using older human insulin. Nobody has measured massage over a rapid-acting analogue or a pump site directly. Studies of local heat on pump sites with rapid-acting insulin point in the same direction, so the safe assumption is that the effect applies.

FIT Canada, the Canadian injection technique guidance, says massaging the injection site immediately before or after an injection is not recommended, because it makes insulin action unpredictable.

What to do

- Don't inject into an area that is about to be worked on.
- Ask the therapist to skip recent injection sites and your pump site.
- No heat — stones, packs, hot towels — over those areas.

How long to keep an area clear depends on the insulin. Your diabetes team can tell you how long yours stays active.

Lows on the table

Lying still, relaxed, eyes closed, face in a cradle: the early signs of a low are easy to miss, or to mistake for relaxation. Diabète Québec warns that blood sugar can drop unexpectedly during massage. No study has looked at lows during massage directly, so what follows is precaution rather than data.

Before. Check your glucose. If you are low or heading down, treat it and wait, or move the appointment. Eat as usual — don't arrive on an empty stomach.

Within reach. Keep fast sugar on the edge of the table or in your hand, not in a bag across the room. Agree on one word with the therapist that means "stop, I need sugar". Many people with type 1 won't interrupt a session they are paying for; this gives you permission in advance.

After. Check again before you drive. The Fédération québécoise des massothérapeutes (FQM) suggests checking before and after at least the first two or three sessions, so you learn how your own numbers respond.

If a low happens during a session

- Stop. Sit up slowly.
- Take 15 g of fast-acting sugar, and recheck in 15 minutes.
- If the person is unconscious or having a seizure, call 911 and give nothing by mouth.

See [treating a low](#) for what 15 g looks like.

Start simple

If you are new to massage with type 1, start with a relaxation massage rather than deep work, and write down your glucose before and after, along with your last dose and where you gave it. Two or three sessions of notes will show your own pattern — which is worth more than any general rule.

Sources: Linde, Diabetes Care 1986 · Koivisto, BMJ 1980 · Koivisto and Felig, NEJM 1978 · Raz et al., Clinical Therapeutics 2009 · Landau et al., Diabetes/Metabolism Research and Reviews 2014 · FIT Canada, Forum for Injection Technique 2020 · Diabète Québec, « Spa, sauna et massothérapie » · FQM, « Masser une personne diabétique » · Diabetes Canada

Sensors, pumps, and when to reschedule

Compression lows, oils and adhesive, keeping your pump within reach, and the few situations where the appointment should wait.

A massage table is one of the few places where someone else's hands spend an hour near your sensor and your infusion set. This page covers what can go wrong with devices during a session, and the short

list of reasons to move the appointment.

Your sensor and pump on the table

Compression lows

Pressure on a CGM sensor — lying on it, or a hand pressing on it — can show a sudden low that isn't real. Confirm a surprising number with a finger prick before acting on it.

In a study where volunteers wore several sensors at once, sudden false readings were strongly linked with lying on the sensors; most of them were drops. The community calls these "compression lows"; the research literature calls them pressure-induced sensor attenuation. Every manufacturer's manual says the same thing: when a reading doesn't match how you feel, check with a meter.

The other direction matters just as much: don't dismiss a real low because you assume it's compression. If in doubt, prick.

Oils and lotions. Dexcom and Abbott both warn against putting lotion near the sensor, because it stops the adhesive holding. Massage oil is the same problem. Neither company says this about pump infusion sets specifically, but the same logic applies. Ask the therapist to leave a hand's width of clear, oil-free skin around each device.

Show where they are. Point out your sensor and infusion set before you lie down. Draping, stretching and friction near them are how they get pulled off.

Pumps. Keep the pump where you can reach it. If you disconnect for the session, agree with your diabetes team how long is fine — and reconnect before you leave.

When to move the appointment

A short list of situations calls for rescheduling:

- **Ketones raised.** Blood ketones of 0.6 or above, or high blood sugar while feeling unwell. That's your sick-day plan, not a massage.
- **A severe low in the last 24 hours** — one that needed someone else's help. Your defences are blunted for the next one.
- **Wounds or infection** in the area to be treated — an open sore, ulcer, blister or infection, especially on the feet.
- **Fever or acute illness.** Sick days come first. Book again once you are back to your usual numbers.

The FQM lists poorly managed diabetes, hypoglycemia episodes and ulcers or wounds as reasons not to massage. Postponing for ketones or a recent severe low is clinical common sense rather than a published massage rule.

Everything else is a reason to talk, not to stay home

Newly diagnosed, pump or injections, existing complications — none of these is a reason to skip massage. They are reasons to tell the therapist.

Some advice circulating online says to get a doctor's clearance before every booking. The FQM suggests checking with your doctor only when in doubt — for example, with circulation problems or complications you are unsure about. Most people with type 1 don't need permission to get a massage.

Sources: Mensh et al., Journal of Diabetes Science and Technology 2013 · Dexcom G7 user guide · Abbott FreeStyle Libre 2 adhesion guide (Canada) · Medtronic Guardian 4 user guide · FQM, « La massothérapie chez la personne diabétique » (2018) · Diabète Québec, « Spa, sauna et massothérapie »

Numb feet, stiff joints and skin

Massage with neuropathy, who to see for a stiff shoulder or hand, lumps at insulin sites, and getting up safely after a session.

Complications don't rule out massage. People with numb feet or stiff shoulders often benefit from hands-on care, at very little risk, as long as the therapist knows what to adjust. This page covers the adjustments.

If your feet or hands have lost feeling

Neuropathy removes the one safety signal massage relies on: you saying "that's too much". So the therapist has to judge for you.

- **Broad pressure.** On numb areas, ask for broad, full-hand pressure rather than deep point work or aggressive stretching.
- **Look afterwards.** Do the same foot check you already do daily again after the session: redness, blisters, broken skin.
- **Draw the map.** Tell the therapist where the numbness starts and stops. That is more useful than the word "neuropathy".

No heat on numb skin

Hot stones, heat packs and hot towels can burn skin that can't feel them. Diabetes Canada already advises testing bath water by hand for the same reason.

There is a small signal of benefit, too. In a trial of 60 people with type 2 diabetes and neuropathy, two weeks of Thai foot massage improved balance and sensation tests. It was small, short and in type 2, so treat it as encouraging rather than proven.

Stiff shoulders and hands: who to see

Frozen shoulder, carpal tunnel, trigger finger and stiff hands are all common after many years with type 1 (see [who does what](#)). Orthotherapists often see them first, because people book for "a stiff shoulder".

Get a diagnosis first. A shoulder that stiffens over weeks, a finger that catches, numb fingers at night: see your doctor or a physiotherapist, not only a massage therapist.

What has the most support. Joint mobilisation combined with exercise, from a physiotherapist or a trained therapist, had the largest effect in a review of eight trials in people with diabetes. The evidence is low quality, and recovery tends to be slower with diabetes — expect it to take a while.

Where massage fits. Alongside, for muscle tension and comfort. Gentle range-of-motion work matters more than force.

If you are offered a steroid injection. A cortisone injection into a shoulder or hand can raise blood sugar for a while afterwards. Plan the days that follow with your diabetes team — see [medicines that raise blood sugar](#).

Lumps, skin and getting up

Lumps where insulin goes. Rubbery lumps under the skin at injection or pump sites — lipohypertrophy — affect about a third of people with type 1, in pooled studies. A therapist may notice them before you do. Insulin given into those spots acts unpredictably, so mention them to your team. It is a site-rotation issue, not a personal failing.

Skin. Avoid any wound, sore, blister or infection in the treatment area. Moisturising dry feet is fine — just not between the toes.

Getting up. Some people who have lived with type 1 for a long time feel light-headed when they stand up. Sit on the edge of the table for a moment, then stand slowly.

Eyes. If you've had recent eye surgery or have advanced retinopathy, ask your eye specialist about time spent face-down.

Sources: Embil et al., Diabetes Canada Clinical Practice Guidelines, foot care chapter 2018 · Chatchawan et al., Medical Science Monitor Basic Research 2015 · Alsubheen et al., Archives of Physical Medicine and Rehabilitation 2019 · Tang and Tan, Medicine 2024 · Deng et al., Journal of Diabetes Investigation 2018 · FQM, « Masser une personne diabétique » (2007)

Before your session: what to say and who to choose

A script to give your therapist, what to bring, how to check a therapist in Québec, insurance receipts, taxes and coverage.

Most of the safety in a massage with type 1 comes from two minutes of conversation before you lie down. This page gives you the words, and explains how to choose a therapist in Québec, where the title alone tells you very little.

What to tell the therapist

Say this, or hand over the [printable therapist card \(PDF\)](#) — two copies on one letter page, with blanks to fill in:

1. I have type 1 diabetes and use insulin — a pump / injections.
2. My sensor is **here**, and my pump site is **here**. No oil, pressure or stretching on them, please.
3. My last insulin was at ___, in my ___. Please skip that area, and no heat there.
4. My glucose right now is ___, and it is steady / rising / falling.
5. If I say "**sugar**", stop — I may be going low.
6. Numb areas: ___. Broad pressure there, and no heat. Also: stiff joints, wounds, or dizziness when I stand up.

Bring your meter or phone, and fast sugar within reach of the table.

People often skip this at the clinic because it feels like a lot to explain to a stranger. Saying it once at home makes it easier — and the card does the explaining for you.

Choosing a therapist in Québec

Check membership. Because massage therapy and orthotherapy aren't regulated in Québec, the title isn't protected: anyone can use it. The association a therapist belongs to is the real check. The Fédération québécoise des massothérapeutes (FQM), Mon Réseau Plus (RMPQ) and RITMA all publish member directories. Complaints go to the therapist's association rather than to a professional order.

One question to ask. "Have you worked with clients who use insulin, a pump or a sensor?" A good answer sounds a lot like the [page for therapists](#).

Insurance, coverage and taxes

Group insurance receipts. Group plans usually require a receipt from a member in good standing of an association that your insurer recognises. The receipt shows the therapist's association and membership number. Some plans also want a doctor's recommendation. Check your insurer's list of recognised associations before you book, not after.

RAMQ. The public plan doesn't cover massage therapy or orthotherapy.

Taxes. In Québec, massage therapy doesn't count for the federal medical expense tax credit — the Canada Revenue Agency recognises massage therapists only in some other provinces. It is generally not eligible for the Québec credit either. Physiotherapy is eligible.

We don't recommend specific clinics or associations. The point is that, in Québec, the patient does the checking that a professional order would do elsewhere.

Where to get help

- **Diabetes Canada:** 1-800-BANTING (1-800-226-8464)
- **Info-Santé:** 811 — a nurse, 24/7, free
- **Emergency:** 911 — unconscious, having a seizure, or won't wake

Every number on these pages is general guidance, not your plan. Take the pattern you record — glucose before and after, last dose and site — to your diabetes team.

Sources: FQM, « Masser une personne diabétique » (2007) · Fédération québécoise des massothérapeutes (FQM) · Mon Réseau Plus (RMPQ) · RITMA · Desjardins Insurance, group health claims guide · Canada Revenue Agency, authorized medical practitioners for the medical expense tax credit (January 2026)

For massage therapists and orthotherapists

An intake checklist, adjustments on the table, and how to recognise and respond to a low in a client with type 1 diabetes.

This page is for the therapist. A client with type 1 diabetes doesn't need a different massage — they need a slightly longer intake, a few adjustments around their devices and insulin sites, and someone in the room who knows what a low looks like.

Intake

Ask, every visit until you know the client:

- **Pump or injections**, and where they last injected or where the pump site is
- **Glucose now, and its direction** — steady, rising or falling
- **Any lows in the last day**, especially one that needed help
- **Numb areas**, wounds or slow-healing sores, dizziness when standing up
- Stiff joints, recent eye surgery, and anything else on your usual health history

The FQM suggests the client check glucose before and after at least the first two or three sessions.

On the table

- **Devices.** Keep a hand's width of clear skin around the CGM sensor and the pump site. No oil there, no pressure, no stretching across them.
- **Insulin sites.** Skip recent injection sites and the pump site, and no heat over them. Massage and heat both make insulin act faster — FIT Canada advises against massaging an injection site right after a dose.
- **Numb skin.** Broad, full-hand pressure rather than deep point work. No hot stones or heat packs on areas the client can't feel.
- **Sugar.** Make sure the client's fast sugar is within *their* reach, not in their bag.
- **Getting up.** Ask them to sit on the edge of the table before standing.

If the sensor alarms during the session, that's not an interruption to apologise for. Give the client a moment to read it. A sudden low reading while they are lying on the sensor may be a compression low — but let them confirm it with a finger prick rather than assuming.

Spot a low

Sweating, trembling, sudden restlessness, confusion, slurred or unusual speech, or simply going quiet. Lying relaxed with eyes closed, these signs are easy to miss.

Sugar first, questions after

- Stop the session. Help them sit up slowly.
- Fast sugar — 15 g — then recheck in 15 minutes.
- Unconscious or having a seizure: call 911. Nothing by mouth.

Things worth noticing

- **Lumps at injection sites.** Rubbery lumps (lipohypertrophy) are common. Insulin acts unpredictably there; a gentle "you might mention that to your diabetes team" is genuinely useful.
- **Stiff shoulders and hands.** Frozen shoulder and stiff hands are common after many years with type 1. If a shoulder has been stiffening for weeks, suggest a diagnosis from a doctor or physiotherapist alongside your work.
- **Reasons to reschedule.** Raised ketones, a severe low in the last 24 hours, fever or acute illness, or wounds and infection in the treatment area.

Most clients with type 1 have explained their diabetes many times. A therapist who asks the right questions first, without fuss, is one they come back to.

Sources: FQM, « Masser une personne diabétique » (2007) · FQM, « La massothérapie chez la personne diabétique » (2018) · Diabète Québec, « Spa, sauna et massothérapie » · FIT Canada 2020 · Dexcom G7 user guide · Diabetes Canada

KRKT's Type 1 Diabetes Education (KRKT:T1DE), from the KRKT Library.

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